

Research paper

Analysis of the ITER central solenoid insert (CSI) coil stability tests

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ABSTRACT

At the end of the test campaign of the ITER Central Solenoid Insert (CSI) coil in 2015, after 16,000 electromagnetic (EM) cycles, some tests were devoted to the study of the conductor stability, through the measurement of the Minimum Quench Energy (MQE). The tests were performed by means of an inductive heater (IH), located in the high-field region of the CSI and wrapped around the conductor. The calorimetric calibration of the IH is presented here, aimed at assessing the energy deposited in the conductor for different values of the IH electrical operating conditions. The MQE of the conductor of the ITER CS module 3L can be estimated as $\sim 200 \text{ J} \pm 20\%$, deposited on the whole conductor on a length of $\sim 10 \text{ cm}$ (the IH length) in $\sim 40 \text{ ms}$, at current and magnetic field conditions relevant for the ITER CS operation. The repartition of the energy deposited in the conductor under the IH is computed to be $\sim 10\%$ in the cable and 90% in the jacket by means of a 3D Finite Elements EM model. It is shown how this repartition implies that the bundle (cable + helium) heat capacity is fully available for stability on the time scale of the tested disturbances. This repartition is used in input to the thermal-hydraulic analysis performed with the 4C code, to assess the capability of the model to accurately reproduce the stability threshold of the conductor. The MQE computed by the code for this disturbance is in good agreement with the measured value, with an underestimation within 15% of the experimental value.

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1. Introduction

While ITER is under construction at Cadarache (France), the ITER Domestic Agencies (DAs) have started the production and delivery of the main components.

In order to ensure the compliance of the Central Solenoid (CS) conductor with the specifications, namely the temperature margin during operation after thousands of cycles, a Central Solenoid Insert (CSI) coil [1,2] has been designed by the US and built in Japan and tested in 2015 in the bore of the Central Solenoid Model Coil (CSMC) [3] at the National Institutes for Quantum and Radiological Science and Technology (former JAEA) in Naka, Japan. Indeed, this is just one of a long series of superconducting (SC) Insert Coils which were previously tested at Naka, starting in 2000, following a very similar procedure [4–7].

The CSI is a circle-in-square Nb_3Sn cable-in-conduit conductor (CICC), cut from the conductor used for the 3L module of the ITER CS and wound in a single-layer solenoid with a hydraulic length of $\sim 43 \text{ m}$. The external side of the jacket is 48.8 mm long, while its

inner diameter is 33 mm ; the two He regions, namely the bundle (where the He flows in the spaces available among the strands) and central channel (also called “hole”) are separated by a spiral with $7 \text{ mm}/9 \text{ mm}$ inner/outer diameter. The other main geometrical characteristics are reported in [8]. It is well instrumented from the electrical, mechanical and thermal-hydraulic point of view, being equipped with several voltage taps, strain sensors, flow meters, pressure taps and thermometers (see the sketch in Fig. 1a for the subset of sensors directly relevant to the present analysis). Moreover, four inductive heaters (IHs), to be used as drivers for the stability and quench tests, are wrapped around the CICC close to the middle of the solenoid, where the magnetic field is maximum. The tests aimed at assessing the conductor performance in operating conditions close to the ITER CS ones, including the check of a possible degradation after thousands of electromagnetic (EM) cycles [2].

At the end of the campaign, some stability and quench tests were performed, with current and magnetic field conditions corresponding to the ITER CS Initial Magnetization (IM, transport current in the CSI $I_{\text{CSI}} = 40 \text{ kA}$ and peak magnetic field $B_{\text{peak}} = 13 \text{ T}$, including background field and self-field) and End of Burn (EoB, $I_{\text{CSI}} = 45.1 \text{ kA}$ and $B_{\text{peak}} = 12.6 \text{ T}$). These are two operating

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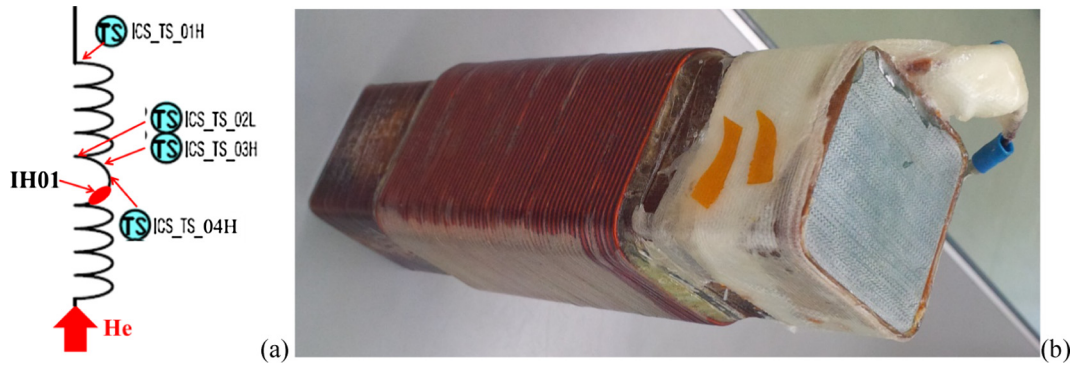


Fig. 1. (a) Sketch of the IH01 and relevant thermometers location in the CSI. (b) Picture of the IH (wound around a dummy support) separately tested in the Naka laboratory and very similar to those installed on the CSI.

conditions of great importance in the 15 MA plasma current operating scenario of the ITER machine as they correspond to the maximum of the magnetic flux density times the transport current in the CS and consequently to the maximum hoop stress. The tests were carried out energizing one IH by AC current pulses with different voltages at the DC power supply, in order to control the energy input; the energy deposited by the IH must be determined by proper calorimetric calibration. Previously, some stability/Minimum Quench Energy (MQE) tests were performed in SULTAN facility on short-samples of similar conductors, but in different operating conditions, namely higher initial temperature (~ 8 K) and lower magnetic field (~ 9 T), not relevant for the ITER CS operation, and for longer (~ 100 ms) heater pulses [9], so that the results are not directly comparable to those obtained for the CSI.

In the present work, the calibration of the IH power deposition with respect to the nominal operating values of its power supply is carried out first. This has been done exploiting both calorimetric data collected in a separate facility on a short conductor sample wrapped with an inductive heater identical to those installed on the CSI [10], and data from heat slug propagation tests performed with the IH on the CSI. The energy fraction deposited separately in the jacket and in the cable is computed by a 3D Finite Elements (FE) EM simulation, describing both the cable and its jacket, suitably tuned to reproduce the global calorimetry of the IH, and confirmed by some features of the available experimental dataset. The computed energy repartition between cable and jacket is then used as an input of the 4C code [11] to assess the MQE of the CSI, both at IM and at EoB, and the latter is compared with the results of the stability tests, analyzed here in detail for the first time on a conductor and in operating conditions relevant for ITER CS coil.

2. CSI stability tests

The IH01 in Fig. 1a, used for the stability and quench tests, consists of 59 turns of 1.6 mm diameter copper wire with insulation, similar to that reported in Fig. 1b, wrapped around the jacket for a length of 10.9 cm. The aim of using an IH instead of a simpler resistive heater is to introduce directly some energy into the cable. However, due to the shape of the IH, which induces eddy currents azimuthally around the conductor axis, determining a shielding effect in the jacket, it is well known that only a very small fraction of the total energy is deposited into the cable [12–15].

The IH01 was energized with a sinusoidal AC current (I) with 1 kHz frequency for 40 ms, in order to simulate a fast disturbance like, e.g., a plasma disruption. The IH power was tuned changing the DC charging voltage (V_{DC}) of the capacitance that, when discharged, feeds the IH circuit, thus modifying the output AC ampli-

tude of both V and I . The energy input to the IH (E_{IH}) may be expressed by the following relation:

$$E_{IH}[J] = k \int I^2 dt \quad (1)$$

where I is the IH AC current evolution during the 40 ms pulse and k is a proportionality constant (to be expressed in Ω).

The experimental data relevant for the IH calibration and stability analysis are those from heat slugs, including tests at ~ 12.5 T background field with no current in the CSI (reported in Table 1), as well as the stability shots (with the CSI charged) reported in Table 2. In the latter, the shots ended with No Quench (NQ), a recovery (R) or a quench (Q). In case of R or Q, the initiation (with or without recovery) of a normal zone with Joule heat generation does not allow to perform a reliable calorimetry, see below.

3. Calibration of the IH

In order to assess the MQE, the experimental IH data must be analyzed in detail to

- Calibrate the dependence of the energy deposited in the conductor on the IH electrical operating conditions (namely, $\int I^2 dt$), by means of calorimetry.
- Assess the fraction of energy deposited in the cable with respect to the total energy deposited in the conductor, by means of a suitable EM model of the IH.

3.1. IH calorimetry

The experimental datasets useful for the IH calibration by calorimetry come from two different sets:

Table 1

Data collected during heat slug tests at ~ 12.5 T background magnetic field (B), with no current in the CSI; E_{cal} is computed according to (2).

$B = 12.5$ T			
Shot#	V_{DC} [V]	$\int I^2 dt$ [$A^2 s$]	E_{cal} [J]
149–2	125	1104	144
149–6	125	1109	150
150–3	140	1391	165
150–4	150	1634	203
150–5	160	1823	237
180–3	160	1810	250
183–4	160	1867	271
192–3	160	1910	233

Table 2

Data collected during stability tests at EoB and IM (NQ = no quench, R = recovery, Q = quench). The current in both CSI and CSMC is also reported for the two testing conditions. E is computed according to (2) for the shots ending with NQ; for the shots ending with R or Q, the energy reported is the E_{IH} as from (3).

EoB ($B_{peak} = 12.6$ T, $I_{CSI} = 45.1$ kA, $I_{CSMC} = 42.8$ kA)				
Shot#	V_{DC} [V]	$\int I^2 dt$ [A ² s]	E [J]	Outcome
181_2	80	499	64.9 ^a	NQ
181_3	100	786	99	NQ
181_4	120	1090	141.7 ^a	NQ
181_5	130	1275	165.8	R
181_6	140	1475	191.6	R
181_7	150	1699	220.9	Q
183_6	140	1472	191.4	Q
184_2	130	1294	177	NQ
184_3	135	1378	179.1	R
184_4	140	1469	191	R
184_5	145	1568	203.8	Q
192_5	135	1410	183.3	Q
IM ($B_{peak} = 13$ T, $I_{CSI} = 40$ kA, $I_{CSMC} = 44.8$ kA)				
Shot#	V_{DC} [V]	$\int I^2 dt$ [A ² s]	E [J]	Outcome
180_5	160	1852	240.8	Q
185_2	80	508	67	NQ
185_3	100	772	103	NQ
185_4	110	929	125	NQ
185_5	120	1094	146	NQ
185_6	130	1257	163.4	R
185_7	140	1454	189	R
185_8	150	1645	213.9	Q

^a Not used for the calorimetric calibration of the IH since the data acquisition was interrupted before the warm He reached TS01H, see Fig. 1a.

- IH shots performed in the CSI without current (shots in Table 1), or with current but without inducing a normal zone in the conductor (NQ shots in Table 2).
- Dedicated tests performed in a separate facility in the Naka laboratory on an IH identical to that installed on the CSI, mounted on a dummy length of conductor.

The total energy deposited by the IH01 installed on the CSI is assessed by means of calorimetry (E_{cal}) using the TS01H temperature sensor (see Fig. 1a), relying on the following assumptions:

- (1) The jacket, cable, bundle He and central channel He have the same temperature below that sensor (this hypothesis is reliable since TS01H thermometer is very far, ~20 m, downstream of the IH).
- (2) The heat transfer to the top stainless steel spacer which, as reported in [1], structurally supports the winding is negligible.

The calorimetry has then been computed as:

$$E_{cal} \sim \int_{t_0}^{t_1} \{\dot{m}_{out}[h_{TS01H}(t) - h_{TS01H}(t_0)]\} dt \quad (2)$$

where $\dot{m}_{out}$ is the outlet mass flow rate, h_{TS01H} is the enthalpy evaluated at temperature $T = TS01H$ (and CSI outlet pressure), t_0 and t_1 are the time integration limits, set according to the heat slug to include the whole energy, see Fig. 2a.

Since the outer surface of the IH can be considered adiabatic, pretty consistently with the fact that it is surrounded by epoxy insulation, E_{IH} includes all the energy deposited in the cable and in the jacket by induction and in the copper wiring by Joule heat generation. The latter will however be conducted to the jacket and then transferred to the He, thus it will be accounted for in the calorimetry.

From the linear best fit of the available set of data coming from the heat slugs in the CSI (including both the dedicated IH calibra-

tion shots of Table 1 and the NQ shots of Table 2, in IM and EoB conditions), we obtain (Fig. 2b).

$$E_{IH}[J] = 0.13[\Omega] \int I^2 dt \quad (3)$$

with an accuracy that can be estimated from the fit R^2 (96.6%) to be ± 25 J (i.e. ~12%) around 1600 A²s. Note that the (small) magnetic field difference between EoB and IM conditions does not affect the k factor.

The best-fit value of k in (1) turns out to be in very good agreement with the results of the dedicated experiment of IH calibration by calorimetry [16,10], where a CS conductor sample, ~20 cm long, was equipped with the IH winding shown in Fig. 1b. The experiment was carried out in a 13 T background field by soaking the conductor sample and the sample holder in liquid He in a cryostat. In this condition, the heat generated by the IH in the sample is absorbed in the liquid He, and then the He is vaporized. A liquid He level sensor was also installed inside the cylinder, so that the vaporized He gas amount was measured by the level sensor as the level change from before to after IH shots. Tests were performed for various AC amplitude of both V and I for the IH windings, resulting in the variation of heat deposited in the sample. Due to the limited volume of the container which collects the He gas, the sample tests have been carried out in a low-energy range. This same procedure was already adopted for the Insert Coils tested in the past [12]. The result of this experiment is reported in Fig. 2b, showing a value of k in (1) of ~0.15 Ω [10], well within the error bar of the best fitting correlation for the data from CSI tests (the green band around the best fit), also considering that the IH sample did not undergo EM cycles.

From the correlation between E_{IH} and $\int I^2 dt$ just developed, it is possible to compute the energy deposited by the IH in the R and Q shots in Table 2, for which a reliable calorimetry could not be performed (see above). This allows the estimation of the MQE, identified as the energy deposited in the shot ending with a quench with the minimum energy deposition, i.e. possibly after a shot with recovery. The shot #180-5 is thus neglected since it was not after tests with recovery, while shot #192-5 should be accounted for

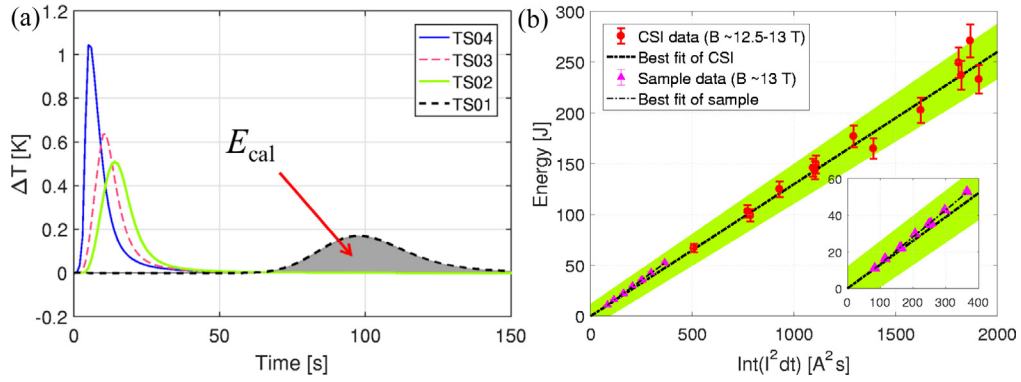


Fig. 2. (a) Typical evolution of temperatures measured downstream the IH during a heat slug; the calorimetric estimate E_{cal} of the energy deposited by the IH in the conductor is proportional to the grey area indicated by the red arrow. (b) Best fit of experimental calorimetry vs. $\int I^2 dt$; the error bar on the experimental data are due to the accuracy of mass flow measurements ($\sim 6\%$), while the error bar on the fit (represented as green band around the fit itself) is estimated by means of the fit R^2 value, combined with the accuracy of the data on which the fit is based, i.e. the above-mentioned accuracy of the mass flow measurements; the data from the independent tests performed on the IH + conductor sample, see below, are also reported (magenta triangles in the bottom-left corner) and shown in the zoom in the inset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

since it shows the lowest quench energy. The measured MQE is, for the shots in EoB conditions, that corresponding to the shot #192-5, i.e. $183 \text{ J} \pm 13\%$, while for the shots in IM conditions it is $214 \text{ J} \pm 11\%$ (shot #185-8). As a result, the MQE can be considered to be $\sim 200 \text{ J} \pm 20\%$ for both IM and EoB conditions, corresponding to $\sim 0.77 \pm 0.15 \text{ J/cm}^3$ of conductor (being the volume of conductor below the heater 260 cm^3), in perfect agreement with the results reported in [10].

3.2. Assessment of the repartition of the energy deposition between jacket and cable

As opposed to what is done in [10], where a simple 1D analytical model assuming a uniform thickness of the jacket is described, a dedicated 3D FE model including the real geometry of the square conductor jacket has been developed here to compute the jacket/cable repartition of the energy deposited by the IH in the conductor. The model, describing the EM behavior of the conductor under the IH, includes the IH itself, the austenitic steel JK2LB jacket region and the cable (inductively coupled); the geometry considered by the model is shown in Fig. 3a, and includes a 30 cm long conductor sample with a 10 cm long IH. Increasing the length of the simulated sample does not change the simulation results.

While in [10] it is modeled as a parallel of copper wires, in the present model the cable is discretized into the 6 petals, as shown in Fig. 3a, each homogenized over its cross section; the electrical contact among neighboring strands within each petal is modeled as an

equivalent conductance ($5 \times 10^7 \text{ S/m}$), calibrated to match the experimental calorimetry on the whole conductor, using as input a perfect sinusoidal current in the IH wires with a root mean square (rms) value obtained equating the computed and experimental $\int I^2 dt$. In view of the fact that the eddy currents are mainly induced in the plane orthogonal to the conductor axis (and of the short length of the heater with respect to the twist pitch of the last cabling stage, $\sim 50 \text{ cm}$), as a first assumption the helical shape of the petal axes has been neglected, assuming a rectilinear configuration. The electrical contact between the jacket and each petal is not accounted for, since a parametric analysis showed that the induced currents will run mostly along azimuthal paths and will not cross the boundary between the jacket and the cable. The fully coupled 3D model developed here can also account for the magnetic coupling between the current induced in the jacket and in the cable (whereas in [10] the magnetic field on the copper wires is computed considering its attenuation due to the jacket through a skin effect approach). After freezing the value of the conductance, see above, the EM model is applied to different stability shots at different power levels, in order to estimate the fraction of energy deposited in the cable. The computed 2D power density distribution over the mid-length cross section of the conductor is presented in Fig. 3b: the power peaks are located close to the central part of the outer edges of the jacket, while the jacket corners are characterized by low eddy current losses, given the large cross section of the eddy currents path at those locations. This level of detail cannot be captured by other 1D simplified models, as the

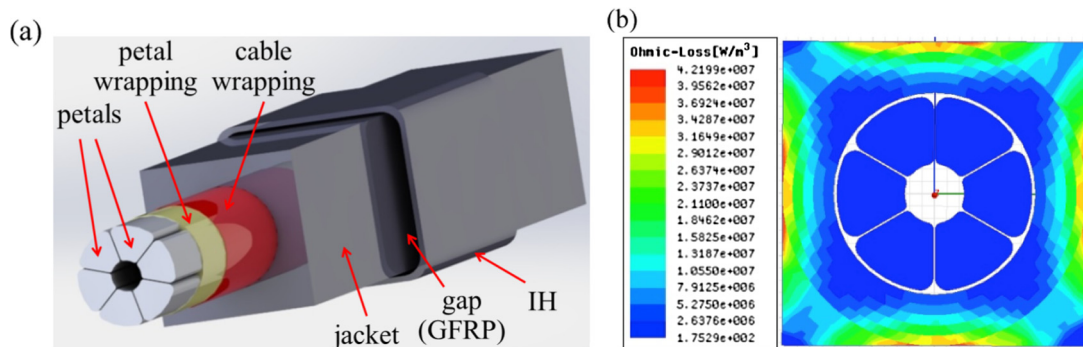


Fig. 3. (a) Geometry of cable and jacket, including petals wrapping, adopted for the calculations of the EM 3D FE model. The gap between the jacket and the IH wires is due to the presence of a GFRP (glass fiber reinforced polymer) spacer with a maximum thickness of 2.5 mm, simulated here as vacuum due to its negligible impact on the magnetic field. (b) Power density distribution in the mid-length conductor cross section at $t = 1.5 \text{ ms}$, for the shot #181-2.

one adopted in [10]. The fraction of energy deposited in the cable computed by the model is $\sim 10\%$ of the total energy, independently on the shot.

Considering a MQE of $200 \text{ J} \pm 20\%$, the corresponding energy deposited in the cable turns out then to be $20 \text{ J} \pm 20\%$, which can be compared with the available heat capacity of the cable bundle. The latter could be estimated as the increase of internal energy ΔU from the initial temperature T_0 up to the current sharing temperature T_{CS} :

$$\Delta U = V_{st} \rho_{st} [c_{st}(T_{CS})T_{CS} - c_{st}(T_0)T_0] + V_{HeB} [\rho_{HeB}(T_{CS})c_{v,HeB}(T_{CS})T_{CS} - \rho_{HeB}(T_0)c_{v,HeB}(T_0)T_0] \quad (4)$$

where V , ρ , c and c_v are the volume (below the IH), density, specific heat and specific heat at constant volume of the strand (index st) and of the bundle He (index He_B), respectively. The additional contribution to the cable stability of ΔH , i.e. the energy which can be removed by the He enthalpy variation during the IH heating time τ_{IH} :

$$\Delta H = +\dot{m}_B [h(T_{CS}) - h(T_0)] \tau_{IH} \quad (5)$$

depends on the He mass flow rate flowing in the bundle $\dot{m}_B$, assumed here to be $\sim 50\%$ of the total (evaluated using the friction factor correlations in [8]). The ΔH contribution can be estimated to be $\sim 10\%$ of the total energy ($\Delta U + \Delta H$), with some uncertainty deriving from the estimation of the mass flow rate in the bundle region.

The available heat capacity of the cable, computed considering the sum of ΔU (4) and ΔH (5), turns out to be $\sim 24 \pm 1 \text{ J}$, fully compatible (within the accuracy) with the energy deposited in the cable, according to the outcome of the FE model.

An indirect confirmation of this estimation comes from a second look at the quench test results, for which the voltage take-off in the voltage taps across the IH is reported in Fig. 4. In the shot #181-7 (EoB), the quench initiation (voltage take-off) occurs right after the end of the heating phase if compared with other shots where the quench is initiated long after the IH pulse is terminated. It is then possible to assume that in the former case the contribution of the jacket heating to the initiation of the normal zone is negligible, and the energy deposited in the cable is sufficient to heat the strands and bundle He at least up to T_{CS} : if the available heat capacity is $\sim 24 \pm 1 \text{ J}$, see (4), and the energy deposited in the conductor under the heater in shot #181-7 is $220 \text{ J} (\pm 12\%)$, the fraction of the energy in the cable is confirmed to be at least $24/220 \sim 10\text{--}11\%$ of the total.

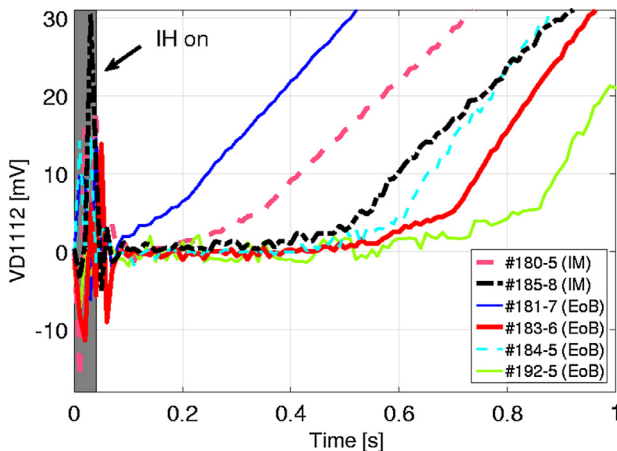


Fig. 4. Evolution of the voltage across the IH during the quench initiation in the stability tests ending with a quench.

Taking into account the energy fraction in the cable, the MQE of the CSI cable corresponds to $\sim 0.4 \text{ J/cm}^3$ of strands (SC and Cu, for a total volume of 53.4 cm^3 below the heater), both in IM and EOB conditions.

Note that the voltage in shot #181-7 takes off before the one in shot #180-5, with a larger energy deposition (see again Table 2), because the former case is in EoB conditions and thus the CSI current is closer to the quench current. For the same initial temperature margin, being the coil current higher, when the T_{CS} is reached also the Joule power generation is higher, so the take-off is anticipated.

4. Analysis of the CSI stability with the 4C code

The thermal-hydraulic code 4C [11], which is being applied to a broad range of transients in the CSI, including the cooldown [8] and the analysis of coupling losses [17], is adopted here to evaluate the MQE both in IM and EoB conditions, using in input the energy repartition between cable and jacket from the EM computation. The input power is imposed constant and equal to the mean value during the IH01 pulse (40 ms). On this heat deposition time scale, the heat can be considered to be deposited in both jacket and strands at the same time, and this is what has been done in the model. The jacket and the cable are directly thermally coupled by means of a heat transfer coefficient of $500 \text{ W/m}^2\text{K}$ [18]. As far as the conductor is concerned, the model includes, as in [13], only the CSI conductor with the two He regions, the strands bundle and the jacket, i.e. no heat transfer to the steel and epoxy spacers is accounted for. Proper constant boundary conditions similar to the experimental conditions are prescribed, namely inlet and outlet pressure and inlet temperature. The geometrical characteristics are reported in Table 3.

Concerning the heat transfer coefficient h_t between the bundle He and the solids (jacket and strands), which are crucial for a proper modeling of the initial fast transient phases, the recipe described in [19] is adopted, which results in an average h_t value of $\sim 5000 \text{ W/m}^2\text{K}$, in agreement with the constant value already used in [13].

A procedure similar to that followed in [13] for the first ITER CSI, and adopted also later by other research units for the stability analysis of the SC outsert coil of the Series Connected Hybrid Magnet System at the National High Magnetic Field Laboratory [20] (where however no comparison with experimental data was possible), has been used here to compute the MQE: several 4C simulations have been performed slightly increasing the input energy, thus allowing to identify the MQE as the first input energy leading to a normal zone propagation and not to a recovery, with an approximation of 0.1 J/cm^3 , i.e. with an uncertainty of the same order of the experimental one. The resulting computed MQE is between 0.6 (the computed MRE) and 0.7 J/cm^3 of conductor, i.e. $\sim 160\text{--}180 \text{ J}$, very close to the experimental one, see Fig. 5. Considering the whole set of stability tests performed, the computed MQE value is almost always within the experimental error bar in EoB conditions, while turns out to slightly underestimate the experiment in IM

Table 3
Geometrical data of the CSI conductor.

Jacket external side	48.8 mm
Jacket internal diameter	33 mm
Void fraction in the annulus	33.6%
Central channel ID/OD	7 mm/9 mm
Total He cross section	285 mm^2
Number of strands (SC + Cu)	576 + 288
Strand diameter	0.83 mm
Cos(θ)	0.954
Wrapping area	30 mm^2

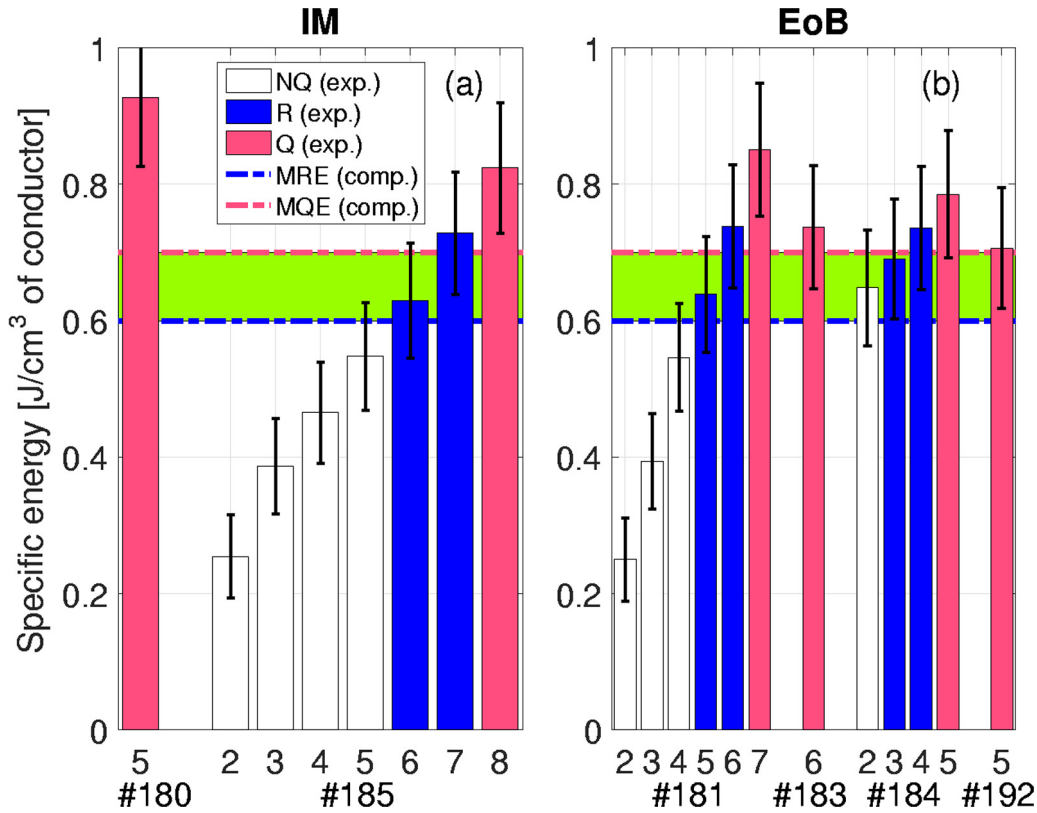


Fig. 5. MQE and Maximum Recovery Energy (MRE) computed by the 4C code, compared to the measured values of the energy deposited in the different stability shots in IM conditions (a), and EoB conditions (b), respectively. The outcome of each shot is depicted with different colors. The error bar is due to the accuracy of the IH calibration (see above).

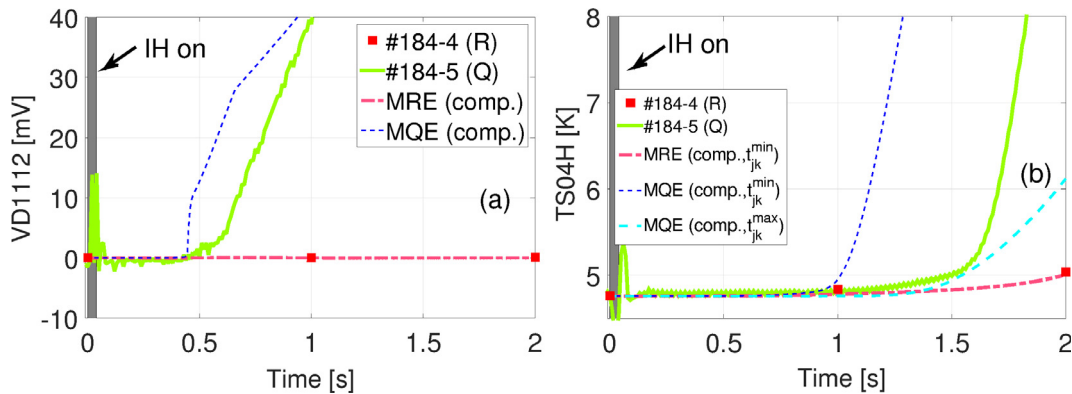


Fig. 6. Comparison between computed and measured evolution of (a) the voltage across the IH01 (VD1112) and (b) the jacket temperature at the TS04 sensor (the closest to the IH01) in the first seconds after the IH01 pulse in EoB conditions. The data of the recovered shot were acquired only at low frequency (1 Hz).

conditions. Also the MRE is in general slightly underestimating the experimental MRE with its error bar, but it should also be noted that, as for the MQE, the differences among different shots are, in absolute value, $\geq 0.1 \text{ J/cm}^3$. The agreement between the 4C calculations and the experiment can then be considered satisfactory. A further reduction of the uncertainty in the computed value is of course possible, but useless in view of the above-mentioned experimental data variation and uncertainty.

The computed MQE is same both in IM and EoB conditions, within the given 20% uncertainty, because the T_{CS} values estimated by the 4C code (in excellent agreement with the measured value) in the two conditions differ only by $\sim 50 \text{ mK}$ [21]. The temperature

margin in the two cases is then the same, thus the same amount of energy will lead to a normal zone propagation or to a recovery.

The evolution of the computed and measured voltage across and jacket temperature few tens of centimeters downstream the IH01 are reported in Fig. 6, for shot #184 in EoB conditions. The simulation at MQE is able to capture the VD1112 take-off time, while the (initial) slope is slightly overestimated as the model does not account for sub-cable current redistributions that can lead to a quench initiation on a single petal instead of on the whole cable [15]. Also the temperature evolution is well captured by the simulation: being the temperature tap stick on the jacket external surface, the heat coming from the bundle He will take some time to

diffuse across the jacket thickness, which is however not constant, in view of the externally square cross section of the CSI jacket. To account for the radial heat diffusion in the jacket, not included in the 4C Mithrandir-like model [22], the jacket surface temperature is obtained post-processing the computed bundle He temperature by means of a 1D radial model, accounting for the jacket thickness t_{jk} and its thermo-physical properties, as already done in [17]. Fig. 6b shows that, depending on the jacket thickness that is adopted in the radial model (the minimum one, corresponding to the jacket edge center, or the maximum, corresponding to the jacket corner), the computed results completely bracket the experimental jacket temperature take-off, confirming the capability of the 4C code to tackle such fast stability transients.

5. Conclusions

The MQE in the CSI was measured during dedicated stability tests in operating conditions (current, magnetic field and temperature margin) relevant for the ITER CS. A short (~ 100 mm) IH was mounted around the conductor to drive fast (40 ms) heating pulses.

In this paper, the detailed analysis of the stability tests performed on the CSI was presented for the first time. The IH calibration via calorimetry was described, allowing the assessment of the energy deposited in the conductor (cable + jacket).

The measured MQE is ~ 200 J $\pm$ 20%, i.e. around 0.77 ± 0.15 J/cm³ (of conductor), both in IM and EoB conditions (~ 12.5 T and 40 and 45.1 kA, respectively). This helps to set an upper bound to the acceptable energy perturbation in the conductor during ITER CS operation: perturbations introducing $\geq \sim 200$ J in ~ 10 cm on a time scale of ~ 10 ms (of the same order of the plasma disruption time scale) must be avoided, as they will likely cause a quench of the magnet.

A suitable FE model of the IH + conductor was used to compute the repartition of the energy deposited inductively by the IH in the jacket and in the cable, the latter being $\sim 10\%$ of the total value as confirmed experimentally by a detailed analysis of the quench initiation in one of the quench tests.

The heat capacity of the whole cable (bundle He + strands) is fully available on the time scale of the thermal disturbances analyzed here and, as such, contributes to the conductor stability.

Finally, the 4C code has been verified to be able to estimate the MQE with an underestimation of the experimental value up to 15%, in both IM and EoB conditions, and can now be used to analyze the previous stability measurements performed for longer (~ 100 ms) perturbations on some ITER CS conductor short samples [9], in order to compare the experimental results obtained there with the stability tests performed on the CSI.

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References

- [1] Isono T et al. Fabrication of an insert to measure performance of iter cs conductor. *IEEE Trans Appl Supercond* 2015;25:4201004.
- [2] Martovetsky N et al. ITER central solenoid insert test results. *IEEE Trans Appl Supercond* 2016;26:4200605.
- [3] Tsuji H et al. Progress of the ITER central solenoid model coil program. *Nucl Fusion* 2001;41:645–51.
- [4] Martovetsky N et al. Test of the ITER central solenoid model coil and CS insert. *IEEE Trans Appl Supercond* 2002;12:600–5.
- [5] Martovetsky N et al. Test of the ITER TF insert and central solenoid model coil. *IEEE Trans Appl Supercond* 2003;13:1441–6.
- [6] Okuno K et al. Test of the NbAl insert and ITER central solenoid model coil. *IEEE Trans Appl Supercond* 2003;13:1437–40.
- [7] Bessette D et al. Test results from the pf conductor insert coil and implications for the ITER PF system. *IEEE Trans Appl Supercond* 2009;19:1525–31.
- [8] Bonifetto R, Brighenti A, Isono T, Martovetsky N, Kawano K, Savoldi L, et al. Analysis of the cooldown of the ITER Central solenoid model coil and insert coil. *Supercond Sci Technol* 2017;30:015015.
- [9] CSJAS SULTAN Sample, Test Report, December 2015.
- [10] Ozeki H, Suwa T, Isono T, Takahashi Y, Nabara Y, Kawano K, et al. Stability test using an inductive heater on the ITER CS insert coil. *J Cryo Super Soc Jpn* 2016;51:113–21.
- [11] Savoldi Richard L, Casella F, Fiori B, Zanino R. The 4C Code for the cryogenic circuit conductor and coil modeling in ITER. *Cryogenics* 2010;50:167–76.
- [12] Matsui K, Nabara Y, Nunoya Y, Koizumi N, Okuno K. Evaluation of inductive heating energy of pf insert coil conductor by calorimetric method. *JAEA-technology report* 85/2008 2009; <http://dx.doi.org/10.11484/JAEA-Technology-2008-085> (date accessed: 08/08/2016).
- [13] Zanino R, Carpaneto E, Portone A, Salpietro E, Savoldi Richard L. Inductively driven transients in the CS Insert Coil (I): heater calibration and conductor stability tests and analysis. *AIP Conf Proc* 2002;613:415–22.
- [14] Savoldi Richard L, Portone A, Zanino R. Tests and analysis of quench propagation in the ITER toroidal field conductor insert. *IEEE Trans Appl Supercond* 2003;13:1412–5.
- [15] Zanino R, Bonifetto R, Savoldi Richard L. Analysis of quench propagation in the ITER poloidal field conductor insert (PFCL). *IEEE Trans Appl Supercond* 2010;20:491–4.
- [16] Ito T et al. Evaluation of inductive heating energy of sub-size improved DPC-C conductor by calorimetric method. *JAERI-Research* 1996;96–042:1–14.
- [17] Breschi M, Bianchi M, Bonifetto R, Carli S, Devred A, Martovetsky N, et al. Analysis of AC losses in the ITER central solenoid insert coil. *IEEE Trans Appl Supercond* 2017;27:4200605.
- [18] Koizumi N, Takeuchi T, Okuno K. Development of advanced Nb3Al superconductors for a fusion demo plant. *Nucl Fusion* 2005;45:431–8.
- [19] Koizumi N, Takahashi Y, Tsuji H. Numerical model using an implicit finite difference algorithm for stability simulation of a cable-in-conduit superconductor. *Cryogenics* 1996;36:649–59.
- [20] Gavrilin AV, Dixon IR, Bonito-Oliva A. Transient stability analysis of the superconducting outsert in the NHMFL Series connected hybrid magnet system. *IEEE Trans Appl Supercond* 2008;18:1280–4.
- [21] Bonifetto R, Isono T, Martovetsky N, Savoldi L, Zanino R. Analysis of the DC performance of the ITER CSI coil using the 4C code. To appear in *Fusion Eng. Design* 2017. <http://dx.doi.org/10.1016/j.fusengdes.2017.02.028>.
- [22] Zanino R, De Palo S, Bottura L. A two-fluid code for the thermohydraulic transient analysis of CI CC superconducting magnets. *J Fus Energy* 1995:14–25.